

Jeewantha Sadaruwan

Electronic and Telecommunication Engineering Undergraduate
University of Moratuwa, Sri Lanka

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PERSONAL STATEMENT

I am an Electronic and Telecommunication Engineering undergraduate focused on Artificial Intelligence, Agentic AI, and Embodied AI. My work centers on developing intelligent systems that can perceive, reason, and act using machine learning, computer vision, large language models, and multi-agent orchestration. I am particularly interested in building AI agents that can use tools, interact with external systems, and support autonomous decision-making in real-world and simulated environments.

Areas of Interest : - Artificial Intelligence, Agentic AI, Large Language Models (LLMs), Multi-Agent Systems, Computer Vision, Embodied AI, Vision-Language Models, and AI Automation.

EDUCATION

University of Moratuwa

B.Sc. in Electronic & Telecommunication Engineering (Third-year undergraduate)

Sri Lanka

Feb. 2023 – Present

- Cumulative GPA: 3.75 / 4.00

Bandaranayake College, Gampaha

G.C.E. Advanced Level Examination 2021 (Physical Science Stream)

Sri Lanka

Jan. 2013 – Jun. 2022

- Z-Score: 2.4875 (3 As-Combined Mathematics, Physics & Chemistry)
- Ranked 166th in the country

ESOFT Metro Campus

Sri Lanka

- Completed Diploma in Information Technology (DITEC) (PEARSON ASSURED) 
- Completed Diploma in English (DIE) (PEARSON ASSURED) 

Certifications

Sri Lanka

- *Supervised Machine Learning: Regression and Classification (Coursera)* *Aug 2024*
 - *Computer Vision (Kaggle)* *Apr 2026*
 - *AI Agents (Hugging Face)* *May 2026*
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EXPERIENCE

WSO2 Lanka (Pvt) Ltd - Robotics & AI Intern

Nov 2025 – Present

- **Project: Unitree Go2 EDU Quadruped Robot - SLAM & Multi-Agent Integration**
 - Integrated a **SLAM based** autonomy pipeline enabling indoor **mapping** and **navigation** on the Unitree Go2 EDU platform.
 - Validated navigation performance across indoor floors with reliable localization and repeatable goal execution.
 - Ongoing: integrating a **multi-agent system** using **RobotecAI RAI** for agent based orchestration and higher-level behavior execution on top of the robotics stack.

AWARDS and SCHOLARSHIPS

- **First Runner - Up IEEE Challenge Sphere Arduino Challenge 2024**  *Oct 2024*
We created a smart organic waste management system that can reduce methane emissions from organic waste as a solution to environmental pollution.
- **First Runner Up - IEEE AI Challenge**  *Oct 2025*
Developed an AI-enabled Smart Transit Companion for trains, providing real-time guidance and commuter assistance.
- **First Runner Up - IEEE Silicon Pulse Competition**  *Feb 2025*
We built a triangular wave function generator using only analog components in 24 hours without any simulation entirely from scratch
- **Finalist - IESL Robogames 2024**  *April 2025*
In the initial rounds, I implemented a maze-solving algorithm in Webots and built a custom robot. In the finals, I controlled a Kobuki robot to complete the challenges.
- **Top 10 - ElectroCombat Battle Robot Competition 2024**  *Jul 2024*
My team and I built a battle robot featuring a vertical spinning weapon system and an efficient control mechanism. Out of over 60 competing teams, we secured a spot in the top 10.
- **Mahapola Higher Education (Merit) Scholarship**
Awarded for outstanding performance in the G.C.E. A/L examination (All-Island Rank 166).

PROJECTS

- ★ **Autonomous Mobile Robot (AMR) Controller for warehouses**  *Mar 2025 – Present*
 - We built an **autonomous robot** for warehouse automation, with real time responsiveness powered by a microcontroller (low level control) and a Jetson Nano B01 (high level control).
 - As part of the project requirements, We developed **custom SLAM algorithm** for this robot and we're currently working on developing warehouse simulation environments in **Gazebo** and implemented the SLAM toolbox and navigation stack using **ROS 2 Humble**.
 - For precise motion control, we used **NEMA 24 stepper motors** with closed-loop drivers, a **Lak-beam 1S LIDAR**, and a **Bosch BNO055 IMU**.

- Fitness Tracker with Machine Learning**  *Aug 2025 – Present*
 - Analyzed accelerometer/gyroscope data from MetaMotion devices collected during gym workouts, visualization, outlier detection, feature engineering, and dimensionality reduction (**PCA**).
 - Built and evaluated ML classifiers (**Naive Bayes, SVM, Random Forest, Neural Network**) to recognize barbell exercise types and estimate repetition counts.

- Ride Booking Agent**  *2025*
 - Developed an AI-powered ride booking system for Sri Lanka that allows users to book rides through natural language conversations using **OpenAI GPT-4o** and **Strands Agents SDK**.
 - Built a full-stack application with **React, TypeScript, FastAPI**, and **PostgreSQL**, including JWT-based authentication, booking management, and persistent user/trip data storage.
 - Integrated agent tools for geocoding, route calculation, real-time vehicle search, dynamic pricing, booking confirmation, and **Google Calendar** synchronization using APIs such as **OpenStreetMap Nominatim** and **OSRM**.

- Vision-Based Navigation of a Kobuki Robot

Jun 2024

 - Developed an autonomous Kobuki robot using Raspberry Pi and OpenCV for real-time obstacle detection and camera-based navigation.
- FingerReader - Smart Assistive Device for the Visually Impaired

Apr 2025

an ESP32-based assistive wearable that reads printed text in real time with speech output, haptic guidance, and fall detection using TinyML for offline OCR/object recognition.

LEADERSHIP EXPERIENCE

- Event Coordinator - Sri Lankan Robotics Challenge (SLRC 2024/2025)

Apr 2024
- Chairman - Sri Lankan Robotics Challenge (SLRC 2025/2026)

Jun 2025 - Present

 - One of Sri Lanka’s largest national robotics competitions, attracting 100 + teams in School category and University category.
 - Led corporate engagement with sponsors and partners

SKILLS SUMMARY

Languages: English (professional proficiency), Sinhala (native proficiency)

Programming Languages: Python, C++, SQL, MATLAB

Machine Learning & Deep Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Neural Networks, TensorFlow, PyTorch, Scikit-learn

LLMs & Agentic AI: Large Language Models, Prompt Engineering, Retrieval-Augmented Generation (RAG), Tool-Using Agents, Multi-Agent Systems, LangChain, LlamaIndex, Strands Agents SDK, OpenAI API

Computer Vision & Perception: OpenCV, Image Processing, Object Detection, Vision-Based Navigation

Data & MLOps Tools: NumPy, Pandas, Matplotlib, Jupyter Notebook, Git, Linux, Docker

Cloud & AI Platforms: AWS, Amazon Bedrock, AWS Lambda, API Integration

Soft Skills: Leadership, Project Management, Critical Thinking, Problem Solving

Interests: Agentic AI, Embodied AI, Vision-Language Models, Robot Learning, AI Automation, Open-Source AI Tools

REFERENCES

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